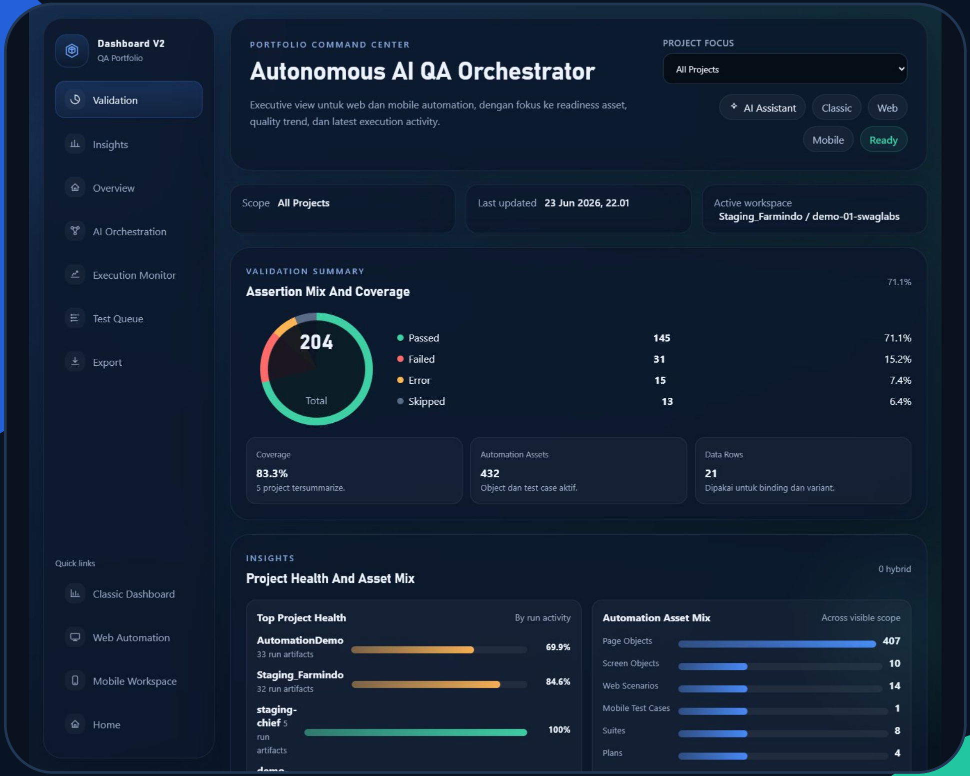
PRIVATE PORTFOLIO CASE STUDY

Autonomous AI QA Orchestrator

Integrated automation workspace for web and mobile quality delivery

Web Automation | Mobile Validation | AI-Assisted Authoring

This document outlines the strategic direction and execution model for the centralized platform I developed. It highlights how the Dashboard V2 validation workspace integrates end-to-end functionality - specifically asset management, content authoring, quality validation, and automated reporting - to streamline complex workflows.



Project Overview: Dashboard V2 Validation Workspace

Problem Statement

Current QA Friction

Before this orchestrator, automation activity was fragmented across scripts, data files, and reporting tools. That slowed release confidence and made it harder to scale shared quality practices.

- Object locators are scattered across script files and become harder to maintain over time.
- Test cases, suites, and datasets live in separate places with limited traceability.
- Crawl, author, run, and result review are not centralized in one operational workspace.
- Scaling across browsers and mobile flows adds setup overhead and fragmented reporting.

What Is Autonomous AI QA Orchestrator?

Project-Based Workspace

Automation assets stay organized by project so object libraries, suites, plans, and run history are easier to track.

The platform behaves as an orchestration layer from setup through reporting, not only as a test runner.

Low-Code Authoring & Auto-Generation

Scenarios can be assembled from reusable page or screen objects to reduce repetitive scripting effort.

The workflow can generate execution-ready assets for Playwright on web and Maestro on mobile.

PLATFORM COVERAGE

Core Capability Map

01

Web Automation

Reusable page objects, browser flows, and validation paths prepared for Playwright-based execution.

02

Mobile Validation

Screen-level objects and mobile scenarios aligned for Maestro-based flow execution and review.

03

Asset Governance

Centralized suites, plans, datasets, and binding assets help teams keep coverage structured and reusable.

04

Execution Visibility

Validation summary, project health, coverage, and artifact review are surfaced in one operational dashboard.

Streamlined Workflow

The workflow reduces context switching by moving teams from discovery to reporting through one repeatable path.

01

Inputs

Web URL, screen flow, or APK target.

→

02

Crawl & Discover

Map UI structure and identify reusable objects.

→

03

Authoring Layer

Compose scenarios from low-code building blocks.

→

04

Execution Engine

Run Playwright for web and Maestro for mobile.

→

05

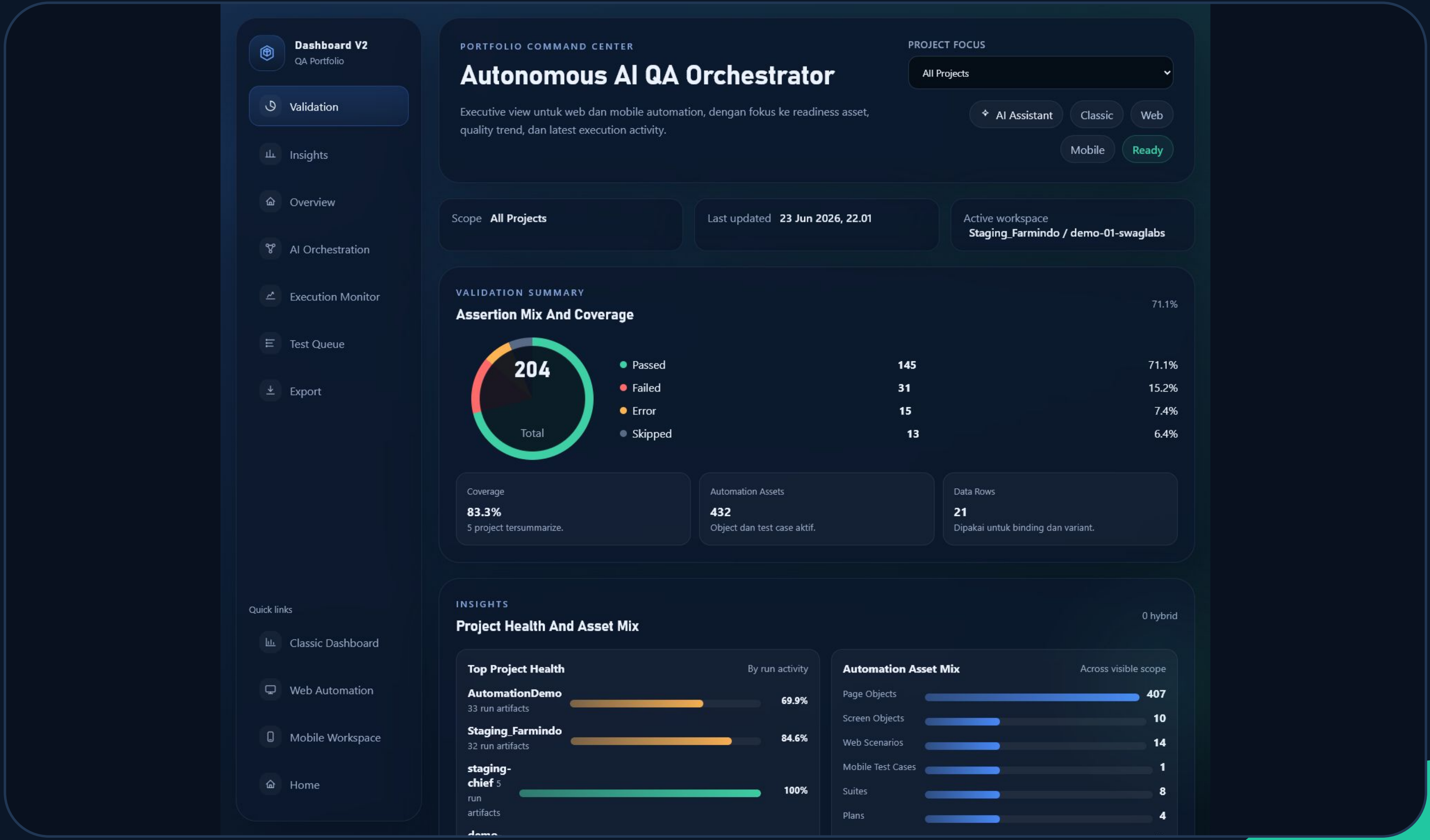
Reports & Artifacts

Collect outcomes, traces, logs, and screenshots.

Implementation Snapshot

A live Dashboard V2 validation view keeps assertion mix, coverage, project health, asset availability, and execution readiness visible in one centralized workspace.

- Validation summary shows pass, fail, error, and skipped distribution at a glance.
- Coverage and asset counters help reviewers understand readiness without opening raw scripts.
- Project health widgets make it easier to compare active workspaces and prioritize follow-up.



Architecture At A Glance

A simple flow keeps discovery, authoring, execution, and reporting connected while preserving reusable QA assets.

01
Inputs
Web URL, screen flow, or APK target.

→

02
Crawl & Discover
Map structure and identify UI objects automatically.

→

03
Authoring Layer
Compose scenarios from object libraries and reusable steps.

→

04
Execution Engine
Playwright for web and Maestro for mobile execution.

→

05
Reports & Insights
Traces, screenshots, metrics, and summary outcomes.

Business Value

This case study is designed to communicate delivery value without exposing the source code while the project is still evolving.

Faster Setup

Reusable discovery and object mapping reduce duplicate effort when new projects start.

Cleaner Maintenance

Shared assets, structured authoring, and centralized visibility help teams control script sprawl.

Private But Reviewable

UI mockups, architecture, and workflow screenshots can be shared safely while code remains private.